EXPERIMENT 10

Applying Performance Optimization Techniques to a Zoho Creator Application

Project: Job Application Management System

Aim

To reduce unnecessary record retrieval and repeated work in the Job Application Management System, review report and form design, and explain suitable approaches to aggregation, batch processing, and indexing in Zoho Creator.

Record Status

This document provides project-specific optimization steps and Deluge examples. Actual Creator before/after timings, schema migrations, and backend index changes must be tested in the live app before they can be reported as completed.

Procedure

1. Identify an operation to optimize. Choose a report or workflow that retrieves Job Application records, such as filtering applications by Job Role or reviewing applications within a date range. Record the dataset size and current result.

2. Use selective criteria. Filter records inside the Creator fetch criteria rather than fetching every application and filtering later in a loop. This reduces unnecessary retrieval when the criteria are selective.

3. Prefer aggregates for totals. When only a count or total is needed, use a report aggregate or a supported aggregate expression instead of looping through every record and manually counting.

4. Review repeated lookups. Avoid repeating the same lookup or calculation inside a loop when a previously retrieved value can be safely reused.

5. Review form and subform design. Keep short, closely related rows in a subform. If an independent history grows over time, consider a separate related form linked with a lookup and filtered report. This is a design recommendation; do not claim a migration unless completed.

6. Use bounded processing. For larger reconciliation or maintenance tasks, process records in controlled batches rather than loading the entire dataset at once. Choose a batch size suitable for Creator limits and verify completeness.

7. Consider indexing with platform support. Do not claim to have created a database index manually. If indexing is needed, consult Zoho Creator support and document any confirmed platform-side changes.

Application Design and Optimization Examples

Technique	Job Application example
Selective fetch	Retrieve only applications for a selected Job Role or status, where such a field exists.
Aggregation	Use a report count for number of applications, or an average Expected Salary by Job Role.
Repeated work	Reuse a calculated value instead of recomputing it for every record.
Form design	Keep the main application form focused on candidate/application details; store independent interview history in a related form if required.
Batch processing	Process large sets of application records in bounded groups when a bulk task is needed.
Indexing	Request platform guidance if a heavily filtered report remains slow; indexing is not claimed as performed.

Deluge Example: Selective Fetch

Example below assumes the form link name is **Job_Application** and a field link name **Job_Role**. Confirm the exact link names in your app before using it. Replace the sample role with a role that exists in your records.

```
selected_role = "Software Developer";
applications = Job_Application[Job_Role == selected_role];
info applications.count();
```

This retrieves records matching the selected role and reports the count. It is a template, not verified against your app's actual link names. If your form or field uses different link names, update them first.

Deluge Example: Filter by Status (Optional)

Use this only if your form actually contains a field named Status and its values match the criterion.

```
applications = Job_Application[Status == "Under Review"];
info applications.count();
```

Validation and Discussion

Check	Validation method	Record result
Correctness	Compare the filtered records with the corresponding Creator report.	Pass / pending
Performance	Run the same query before and after with comparable data and record the time.	Pending until measured
Analytics	Reconcile counts and salary averages with source records.	Verified / pending
Batch completeness	Confirm all intended records are processed exactly once.	Verified / pending
Indexing	Only record indexing if Zoho confirms a platform-side change.	Not performed unless confirmed

Result

The project-specific performance optimization plan identifies selective record retrieval, aggregation, reduced repeated work, focused form design, and bounded batch processing as applicable techniques. Live performance improvements are not claimed until the same operation is measured before and after in Zoho Creator.

Viva Focus

Be prepared to explain why filtering during retrieval is preferable to fetching all records, when aggregates are useful, why large independent histories may belong in a related form, and why local algorithm timings must not be presented as Creator platform measurements.

Note: The source reference discusses claim records. This version adapts the same optimization concepts to the Job Application Management System and uses illustrative Deluge examples; verify field link names and test in the live app.